

Applications of Stochastic Calculus

1 Foundations and Definitions

1.0.1 Discrete-Time Martingale

A sequence $(M_n)_{n \in \mathbb{N}}$ adapted to a filtration $(\mathcal{F}_n)_{n \in \mathbb{N}}$ is a martingale if it is square-integrable and satisfies :

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{for all } n \in \mathbb{N}.$$

Any sum of centered independent increments forms a martingale : if $\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = 0$, then $(Y_0 + \dots + Y_n)$ is a martingale with respect to its natural filtration.

1.0.2 Continuous-Time Martingale

A continuous-time process $(M_t)_{t \geq 0}$ adapted to $(\mathcal{F}_t)_{t \geq 0}$ with $\mathbb{E}[|M_t|] < \infty$ is a martingale if :

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{for all } 0 \leq s \leq t.$$

Every integrable process with centered independent increments is a martingale with respect to its natural filtration. In particular, for any square-integrable random variable F , the process $(\mathbb{E}[F \mid \mathcal{F}_t])_{t \geq 0}$ is always a martingale by the tower property.

2 Types of Martingales in the Textbook

2.1 The Itô Stochastic Integral

For any $u \in L^2_{ad}(\Omega \times \mathbb{R}_+)$, the indefinite Itô integral is a martingale :

$$\mathbb{E} \left[\int_0^t u_s dB_s \mid \mathcal{F}_r \right] = \int_0^r u_s dB_s \quad \text{for all } 0 \leq r \leq t.$$

This follows from Corollary 1.1 and is the bedrock of all subsequent martingale constructions. In particular, the Itô integral of a deterministic function $f \in L^2(\mathbb{R}_+)$ produces a Gaussian random variable with mean zero and variance $\int |f(t)|^2 dt$.

$$\mathbb{E} \left[\int_0^\infty u_s dB_s \right] = 0, \quad \mathbb{E} \left[\left(\int_0^\infty u_s dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^\infty |u_s|^2 ds \right].$$

This is the Itô isometry.

2.2 The Discounted Price Process (Risk-Neutral Martingale)

The core martingale of Black-Scholes and all bond pricing models. Under the risk-neutral measure Q , the discounted asset price is a martingale :

$$\tilde{S}_t = \exp \left(- \int_0^t r_s ds \right) S_t \quad \text{is an } \mathcal{F}_t\text{-martingale under } Q.$$

For self-financing portfolios, the discounted portfolio value also becomes a stochastic integral, hence a martingale, under Q :

$$\tilde{V}_t = \tilde{V}_0 + \int_0^t \sigma_u \eta_u \tilde{S}_u d\tilde{B}_u.$$

This is the fundamental link between self-financing portfolios and martingales : a portfolio is self-financing if and only if its discounted value is a stochastic integral martingale under Q .

2.3 The Discounted Bond Price Process

Under the absence of arbitrage, for any maturity T , the discounted zero-coupon bond price is an $\mathcal{F}_t$ -martingale under P :

$$t \mapsto \exp \left(- \int_0^t r_s ds \right) P(t, T) = \mathbb{E}_Q \left[\exp \left(- \int_0^T r_s ds \right) \mid \mathcal{F}_t \right].$$

This is not assumed but derived : since $P(t, T)$ is defined as the conditional expectation of the discounted payoff, the tower property of conditional expectations immediately gives the martingale property. This is the key reason why bond pricing can be formulated as a PDE – the dt terms in the Itô expansion must vanish.

The SDE of the discounted bond price is purely driven by $d\tilde{B}_t$:

$$d \left[e^{-\int^r P(t, T)} \right] = e^{-\int^r \sigma(t, r_t)} \frac{\partial F}{\partial x}(t, r_t) d\tilde{B}_t.$$

2.4 The Forward Measure Density Process

The Radon-Nikodym process connecting P to the forward measure $\tilde{P}$ is itself a martingale :

$$\Psi(t) = \mathbb{E}_P \left[\frac{d\tilde{P}}{dP} \mid \mathcal{F}_t \right] = \frac{P(t, T)}{P(0, T)} \exp \left(- \int_0^t r_s ds \right).$$

This satisfies the SDE $d\Psi(t) = \Psi(t) \zeta_t dB_t$, where $\zeta_t = \sigma \partial \log F / \partial x$ is the bond volatility. Solving this gives the Girsanov density explicitly as an exponential martingale.

2.5 Bond Price Ratios (Proposition 7.2)

For any two maturities $S, T \geq 0$, the ratio of bond prices is a martingale under the forward measure $\tilde{P}$ with maturity T :

$$t \mapsto \frac{P(t, S)}{P(t, T)} \quad \text{is an } \mathcal{F}_t\text{-martingale under } \tilde{P}.$$

Its SDE under $\tilde{P}$ is purely driven by the Brownian motion $\tilde{B}_t$:

$$d \left[\frac{P(t, S)}{P(t, T)} \right] = \frac{P(t, S)}{P(t, T)} \left(\zeta^S(t) - \zeta^T(t) \right) d\tilde{B}_t.$$

This is used extensively in the pricing of bond options and swaptions : the forward price $P(t, S)/P(t, T)$ is the natural underlying under $\tilde{P}$.

2.6 The Annuity Numeraire (Proposition 10.3)

The discounted annuity numeraire is a martingale under P :

$$t \mapsto \exp \left(- \int_0^t r_s ds \right) P(t, T_i, T_j) \quad \text{is a martingale under } P.$$

where

$$P(t, T_i, T_j) = \sum_{k=i}^{j-1} \delta_k P(t, T_{k+1})$$

is the sum of discounted bond prices weighted by interval lengths. This result follows by linearity from each discounted bond being a martingale.

2.7 The Forward Swap Rate (Proposition 10.8)

The forward swap rate $S(t, T_i, T_j)$ is a martingale under the forward swap measure $P_{i,j}$:

$$dS(t, T_i, T_j) = S(t, T_i, T_j) \sigma_{i,j}(t) dB_t^{i,j}.$$

This is the mathematical foundation of the Black formula for swaption pricing. The swap rate volatility $\sigma_{i,j}(t)$ involves the bond volatilities $\zeta_k(t)$ and the numeraire weights

$$v_k^{i,j}(t) = \frac{P(t, T_k)}{P(t, T_i, T_j)}.$$

2.8 The Forward LIBOR Rate (BGM Model)

In the BGM model, each forward LIBOR rate $L(t, T_i, T_{i+1})$ is modeled as a geometric Brownian motion, hence a martingale, under the forward measure P_{i+1} :

$$\frac{dL(t, T_i, T_{i+1})}{L(t, T_i, T_{i+1})} = \gamma_i(t) dB_t^{i+1}.$$

This positivity-preserving choice is why the BGM model is practically favored : it yields Black-Scholes caplet pricing formulas directly, while under the terminal measure P_n the dynamics include a nonlinear drift correction.

2.9 The Inverse Density Process (Section 7.4)

The inverse forward measure density is also a martingale under $\tilde{P}$:

$$t \mapsto \frac{1}{P(t, T)} \exp \left(\int_0^t r_s ds \right) \quad \text{is an } \mathcal{F}_t\text{-martingale under } \tilde{P}.$$

This result (Proposition 7.4) is used in swaption pricing when computing expectations under $\tilde{P}$ that involve the inverse of bond prices or numeraires.

3 Key Properties of Martingales

3.1 Tower Property (Iterated Conditioning)

For any sub- σ -algebras $\mathcal{G} \subset \mathcal{F}$ and any integrable F :

$$\mathbb{E}[\mathbb{E}[F \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[F \mid \mathcal{F}_s] \quad \text{for all } 0 \leq s \leq t.$$

This is used constantly : to show that a conditional expectation process is a martingale, to collapse nested expectations when computing bond prices, and in the key identity

$$e^{-\int^r P(t, T)} = \mathbb{E}[e^{-\int_0^T r} \mid \mathcal{F}_t].$$

It underlies Propositions 7.1, 7.2, 10.3, and the entire forward measure machinery.

3.2 Vanishing of dt Terms

A crucial characterization (Corollary of Protter) : if a continuous semimartingale Y_t is a martingale, then its Itô decomposition has no finite variation (dt) term – only a local martingale (dB) part survives. This means :

$$\text{If } t \mapsto Y_t \text{ is a martingale, then in } dY_t = \alpha_t dt + \beta_t dB_t, \quad \alpha_t = 0.$$

This is the technique used in Chapter 4 to derive the bond pricing PDE and in Chapter 6 to derive the HJM condition. It is also used in Chapters 7 and 10 whenever bond dynamics under a new measure need to be determined.

3.3 Uniqueness of Martingale Measures

When the martingale measure is unique, the market is complete (no arbitrage, perfect hedging). The Girsanov theorem guarantees the existence of such a measure by adjusting the drift of Brownian motion. The market price of risk K_t parametrizes the family of equivalent martingale measures.

3.4 Predictable Representation Property

In Brownian filtrations, every martingale M_t can be represented as a stochastic integral :

$$F = \mathbb{E}_Q[F] + \int_0^T \xi_t d\tilde{B}_t \quad \text{for some adapted process } (\xi_t).$$

This is used in Proposition 2.2 and Lemma 2.2 to construct the Delta-hedging strategy : the integrand

$$\xi_t = \sigma_t S_t \frac{\partial}{\partial x}(P_{t,T} \phi)(S_t)$$

directly gives the number of shares to hold in the replicating portfolio.

4 Core Techniques and Tricks

4.1 Technique 1 : Girsanov's Change of Measure

The Girsanov theorem is the fundamental tool for converting between different martingale measures. If (ψ_t) satisfies the Novikov condition

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int |\psi_t|^2 dt \right) \right] < \infty,$$

then defining

$$\frac{dQ}{dP} = \exp \left(- \int_0^T \psi_s dB_s - \frac{1}{2} \int_0^T \psi_s^2 ds \right)$$

makes

$$\hat{B}_t = B_t + \int_0^t \psi_s ds$$

a Q -Brownian motion.

How it is used : (1) In Black-Scholes, $\psi_t = (\mu_t - r_t)/\sigma_t$ removes the drift to make $\tilde{S}_t$ a Q -martingale. (2) For bond options (Section 7.2), $\psi_t = -\zeta_t$ (the bond volatility) constructs the forward measure $\tilde{P}$ such that $P(t, S)/P(t, T)$ becomes a $\tilde{P}$ -martingale. (3) In Chapter 8, the same technique constructs the

Hull-White forward measure. The Girsanov density always satisfies $d\Psi = \Psi\psi_t dB_t$, i.e. it is itself an exponential martingale.

4.2 Technique 2 : The Vanishing- dt Argument (Martingale Extraction)

This is the most frequently used algebraic trick in the textbook. The procedure is : (1) Write down an expression X_t that you know is a martingale from economic arguments. (2) Apply Itô's formula to obtain $dX_t = (dt \text{ terms}) + (dB \text{ terms})$. (3) Since X_t is a martingale, all dt terms must equal zero by the characterization of continuous martingales. (4) Setting the coefficient of dt to zero yields the desired equation.

Applications : Setting coefficient of $dt = 0$ in $d[e^{-\int^r P(t,T)}] = 0$ gives the bond pricing PDE. Setting coefficient of $dt = 0$ in $d[e^{-\int^r e^{-X_t}}] = 0$ gives the HJM no-arbitrage condition

$$\int_t^T \alpha(t,s)ds = \frac{1}{2} \left(\int_t^T \sigma(t,s)ds \right)^2,$$

which upon differentiation in T yields

$$\alpha(t,T) = \sigma(t,T) \int_t^T \sigma(t,s)ds.$$

4.3 Technique 3 : Change of Numeraire

Pricing formula (Proposition 7.1) : For any integrable payoff F at time T ,

$$\mathbb{E}_P \left[e^{-\int_t^T r_s ds} F \mid \mathcal{F}_t \right] = P(t,T) \mathbb{E}_{\tilde{P}}[F \mid \mathcal{F}_t].$$

The trick is to absorb the stochastic discount factor into the measure. This isolates the payoff F from the interest rate randomness, enabling pricing by computing a simpler expectation under $\tilde{P}$. For bond options, $F = (P(T,S) - K)^+$ and $P(T,S) = P(t,S)/P(t,T) \cdot \exp(\text{Gaussian})$ under $\tilde{P}$, reducing to the Black-Scholes formula.

For swaptions (Section 10.7) :

$$\mathbb{E}_P \left[e^{-\int^r (S(T_i, T_i, T_j) - \kappa)^+} \right] = P(t, T_i, T_j) \mathbb{E}_{P_{i,j}} [(S(T_i, T_i, T_j) - \kappa)^+ \mid \mathcal{F}_t],$$

using the annuity numeraire.

4.4 Technique 4 : Conditional Expectation as Martingale

The most direct way to produce a martingale : for any integrable F , the process $M_t = \mathbb{E}[F \mid \mathcal{F}_t]$ is always an $\mathcal{F}_t$ -martingale. This is used to define bond prices, the forward measure density, and conditional survival probabilities in credit models.

Combined with the Markov property, this allows replacing conditioning on $\mathcal{F}_t$ with conditioning on the current state r_t alone, so that $P(t,T) = F(t,r_t)$ becomes a deterministic function – enabling the PDE approach.

4.5 Technique 5 : Freezing and Approximation

In the BGM model (Section 11.3), the swap rate volatility $\sigma_{i,j}(t)$ involves stochastic quantities (LIBOR rates, bond price ratios). These are “frozen” at their time- t values to turn $\sigma_{i,j}$ into a deterministic function, so that $S(t, T_i, T_j)$ becomes approximately log-normal under $P_{i,j}$. This approximation is justified because $S(t, T_i, T_j)$ is a martingale under $P_{i,j}$ regardless – the approximation only concerns its distributional law (log-normality), not the martingale property itself.

4.6 Technique 6 : Itô's Product Rule / Numeraire Invariance

When pricing and hedging involve ratios of prices, e.g. $V_t/P(t,T)$, Itô's product rule is applied to compute dV_t from $d(P(t,T)\hat{C}_t)$:

$$dV_t = d(P(t, T_i, T_j)\hat{C}_t) = \hat{C}_t dP + P d\hat{C}_t + dP d\hat{C}_t.$$

The cross term $dP d\hat{C}_t$ is computed using the quadratic covariation rules. This technique recovers the self-financing condition from the martingale property of the forward portfolio, and explicitly gives the hedging coefficients ξ_t^T and ξ_t^S in Exercises 7.4 and 10.4.

5 Where Martingales Are Used

5.1 Bond Pricing PDE (Chapter 4)

The martingale property of $e^{-\int^r P(t,T)}$ is the single fact from which the entire bond pricing PDE is derived. Applying Itô to $F(t, r_t)$ and demanding the dt coefficient vanish gives :

$$\frac{\partial F}{\partial t} + \tilde{\mu}(t,x) \frac{\partial F}{\partial x} + \frac{1}{2} \sigma^2(t,x) \frac{\partial^2 F}{\partial x^2} = xF, \quad F(T,x) = 1.$$

In the Vasicek model this yields the affine solution $F(t,x) = \exp(A(T-t) + xC(T-t))$ via a system of Riccati ODEs. The probabilistic method confirms this using the Gaussian characteristic function of the stochastic integral.

5.2 HJM No-Arbitrage Condition (Chapter 6)

The process

$$\exp \left(- \int_0^t r_s ds \right) P(t,T) = \exp \left(- \int_0^t r_s ds - X_t \right)$$

must be a Q -martingale. Applying Itô and setting the dt coefficient to zero yields the HJM condition :

$$\alpha(t,T) = \sigma(t,T) \int_t^T \sigma(t,s) ds.$$

This is verified in the Vasicek model where $\sigma(s,t) = \sigma e^{-b(t-s)}$ and

$$\alpha(t,T) = \sigma^2 e^{-b(T-t)} \int_t^T e^{-b(s-t)} ds.$$

The Markov property of short rates in HJM holds when $\sigma(s,t) = \xi(s)\psi(t)$ – the product condition – from which the Hull-White SDE follows.

5.3 Derivative Pricing via Forward Measure (Chapter 7)

Bond call option pricing uses three martingale results simultaneously : (1) $P(t,S)/P(t,T)$ is a $\tilde{P}$ -martingale, giving $\mathbb{E}_{\tilde{P}}[P(T,S) \mid \mathcal{F}_t] = P(t,S)/P(t,T)$. (2) $P(T,S)$ is log-normal under $\tilde{P}$ with Gaussian exponent X . (3) The log-normal expectation formula (Lemma 2.3) gives the closed-form bond option price :

$$\mathbb{E}_P \left[e^{-\int^r (P(T,S) - K)^+ \mid \mathcal{F}_t} \right] = P(t,S) \Phi(d_+) - K P(t,T) \Phi(d_-),$$

where

$$d_{\pm} = \frac{1}{v} \log \left(\frac{P(t,S)}{K P(t,T)} \right) \pm \frac{v}{2}, \quad v^2 = \int_t^T |\zeta^S(s) - \zeta^T(s)|^2 ds.$$

5.4 Caplet Pricing (Chapters 7 and 11)

Caplet pricing under the T_{k+1} -forward measure :

$$\mathbb{E} \left[e^{-\int^r (f(T_k, T_k, T_{k+1}) - \kappa)^+ \mid \mathcal{F}_t} \right] = P(t, T_{k+1}) \mathbb{E}_{k+1} [(L(T_k, T_k, T_{k+1}) - \kappa)^+ \mid \mathcal{F}_t]$$

In the BGM model, $L(t, T_k, T_{k+1})$ is a geometric Brownian motion (martingale) under P_{k+1} , so the inner expectation is exactly the Black-Scholes formula

$$Bl(\kappa, L(t, T_k, T_{k+1}), \sigma_k(t), 0, T_k - t)$$

with

$$\sigma_k^2(t) = (T_k - t)^{-1} \int_t^{T_k} |\gamma_k(s)|^2 ds.$$

5.5 Credit Default Modeling (Chapter 9)

The survival probability process uses the martingale property of conditional expectations : for any $\mathcal{F}_T$ -measurable F , Lemma 9.1 gives :

$$\mathbb{E}[F \mathbf{1}_{\{\tau > T\}} \mid \mathcal{G}_t] = \mathbf{1}_{\{\tau > t\}} \mathbb{E} \left[F \exp \left(- \int_t^T \lambda_u du \right) \mid \mathcal{F}_t \right].$$

This reduces defaultable bond pricing to the same martingale framework as default-free bonds, with the short rate r_t replaced by $r_t + \lambda_t$ (rate plus default intensity). The key insight is that conditioning on $\{\tau > t\}$ converts the problem from the enlarged filtration $\mathcal{G}_t$ back to $\mathcal{F}_t$.

5.6 Swap Rate Dynamics and Swaption Pricing (Chapters 10–11)

The swap rate

$$S(t, T_i, T_j) = \frac{P(t, T_i) - P(t, T_j)}{P(t, T_i, T_j)}$$

is a martingale under $P_{i,j}$ (it equals $v_i^{i,j}(t) - v_j^{i,j}(t)$, each of which is a martingale). Its SDE under $B^{i,j}$ is geometric, which – together with the approximation that freezes random coefficients – gives the swaption approximation formula :

$$\text{Swaption price} \approx P(t, T_i, T_j) Bl(\kappa, S(t, T_i, T_j), \tilde{\sigma}_{i,j}(t), 0, T_i - t).$$

The BGM calibration then uses this formula inversely : given market swaption prices, infer the LIBOR volatility functions $\gamma_k(t)$ by minimizing the RMS distance between implied and model swaption volatilities.

6 Problem-Solving Techniques in Exercises

6.1 Computing $P(t,T)$: The Gaussian Trick

When

$$r_t = g(t) + \int_0^t h(t,s) dB_s$$

is Gaussian (Vasicek, Ho-Lee), compute $P(t,T) = \mathbb{E}[e^{-\int^r} \mid \mathcal{F}_t]$ by : (1) Separate the known part ($\mathcal{F}_t$ -measurable) from the future stochastic integral. (2) Apply the Gaussian Laplace transform

$$\mathbb{E}[e^{\alpha X}] = e^{\alpha \mu + \alpha^2 \sigma^2 / 2}.$$

(3) The result is an affine exponential in r_t : $P(t,T) = \exp(A(T-t) + r_t C(T-t))$.

6.2 Verifying the Martingale Property

Standard procedure : (1) Compute dX_t by Itô's formula. (2) If $dX_t = \beta_t dB_t$ (no dt term), then X_t is a local martingale. (3) Verify integrability to confirm it is a true martingale. Alternatively, if $X_t = \mathbb{E}[Y \mid \mathcal{F}_t]$ for some integrable Y , the martingale property is immediate from the tower property.

6.3 Change of Measure in Exercises

Step 1 : Find $d\Psi(t) = \Psi(t)\zeta_t dB_t$ using the martingale property of the discounted bond. Step 2 : Solve this SDE to get

$$\Psi(T) = \exp \left(\int \zeta_s dB_s - \frac{1}{2} \int |\zeta_s|^2 ds \right).$$

Step 3 : Apply Girsanov :

$$\tilde{B}_t = B_t - \int_0^t \zeta_s ds$$

is a Brownian motion under $\tilde{P}$. Step 4 : Rewrite all SDEs in terms of $d\tilde{B}_t$ to get dynamics under $\tilde{P}$.

6.4 Option Pricing by Log-Normal Expectation (Lemma 2.3)

Whenever a payoff is of the form $(e^{m+X} - K)^+$ with $X \sim N(0, v^2)$:

$$\mathbb{E}[(e^{m+X} - K)^+] = e^{m+v^2/2} \Phi \left(v + \frac{m - \log K}{v} \right) - K \Phi \left(\frac{m - \log K}{v} \right).$$

The mean m is always identified using the martingale property : since

$$\frac{P(t,S)}{P(t,T)} = \mathbb{E}_{\tilde{P}}[P(T,S) \mid \mathcal{F}_t] = \mathbb{E}[e^{m+X} \mid \mathcal{F}_t] = e^{m+v^2/2},$$

one obtains $m + v^2/2 = \log(P(t,S)/P(t,T))$ without any explicit calculation.

6.5 The Conditional Variance Shortcut

For stochastic integrals of the form $\int_t^T f(T-s) dB_s$ that are independent of $\mathcal{F}_t$, the variance is computed directly by the Itô isometry without conditioning :

$$\text{Var} \left[\int_t^T f(T-s) dB_s \mid \mathcal{F}_t \right] = \int_t^T |f(T-s)|^2 ds.$$

The term $\int_0^t f(T-s)dB_s$ is $\mathcal{F}_t$ -measurable and contributes only to the conditional mean, not the conditional variance. This separation is used in Exercise 4.5 and Problem 8.2 to compute bond prices in affine models.

7 Quick Reference : Martingale Summary

- Itô stochastic integral**
Form : $\mathbb{E}[\int u \, dB \mid \mathcal{F}_t] = \int_0^t u \, dB$
Use : Pricing, hedging, PDE derivation
- Discounted asset price** $\tilde{S}_t$
Form : $\tilde{S}_t = e^{-\int^r} S_t$ under Q
Use : No-arbitrage condition, Black-Scholes
- Discounted bond** $e^{-\int^r} P(t,T)$
Form : $\mathbb{E}_Q[e^{-\int_0^T r \mid \mathcal{F}_t}]$
Use : Bond pricing PDE, HJM condition
- Girsanov density** $\Psi(t)$
Form : $P(t,T)P(0,T)^{-1}e^{-\int^r}$
Use : Defines forward measure $\tilde{P}$
- Bond ratio** $P(t,S)/P(t,T)$
Form : Under $\tilde{P}$ with numeraire $P(t,T)$
Use : Bond option pricing
- Discounted annuity** $e^{-\int^r} P(t,T_i,T_j)$
Form : Under P by linearity
Use : Swaption pricing framework
- Swap rate** $S(t,T_i,T_j)$
Form : Under forward swap measure $P_{i,j}$
Use : Swaption Black formula
- LIBOR rate** $L(t,T_i,T_{i+1})$
Form : Geometric BM under P_{i+1}
Use : Cap pricing, BGM model
- Inverse density** $P(t,T)^{-1}e^{\int^r}$
Form : Under $\tilde{P}$
Use : Swaption pricing section 7.4
- Conditional survival** $\mathbf{1}_{\{\tau>t\}}\mathbb{E}[e^{-\int^\lambda} \mid \mathcal{F}_t]$
Form : Under $\mathcal{G}_t$ via Lemma 9.1
Use : Defaultable bond pricing

Note : All martingale properties in this text ultimately rest on two foundational results – the Itô isometry (which makes stochastic integrals L^2 -isometries and hence martingales) and the Girsanov theorem (which makes drift removal and measure changes explicit). Every pricing formula, every PDE, and every no-arbitrage condition in stochastic interest rate modeling is a consequence of applying these two results together with the tower property of conditional expectations.

8 Financial Instruments

8.1 Zero-Coupon Bond

A zero-coupon bond is the most fundamental instrument in interest rate theory. It is a contract priced $P(t,T)$ at time $t < T$ that delivers \$1 with certainty at maturity T , so $P(T,T) = 1$.

Deterministic rate. When the short rate $r > 0$ is constant,

$$P(t,T) = e^{-r(T-t)}.$$

When $(r_s)_{s\geq 0}$ is a deterministic function of time,

$$P(t,T) = \exp\left(-\int_t^T r_s \, ds\right).$$

Stochastic rate. Under the risk-neutral measure Q , the arbitrage-free bond

price is the conditional expectation

$$P(t,T) = \mathbb{E}^Q\left[\exp\left(-\int_t^T r_s \, ds\right)\middle|\mathcal{F}_t\right].$$

This formula is the central equation of bond pricing. The discount factor $\exp(-\int_t^T r_s \, ds)$ is random because (r_s) is a stochastic process.

Coupon bond. A bond with continuous coupon rate $c > 0$ has price

$$P_c(t,T) = \int_t^T c \, P(t,s) \, ds + P(t,T)$$

Defaultable bond. When default can occur at a random time τ with intensity process (λ_t) , the zero-recovery bond price satisfies

$$P(t,T) = \mathbf{1}_{\{\tau>t\}} \mathbb{E}\left[\exp\left(-\int_t^T (r_s + \lambda_s) \, ds\right)\middle|\mathcal{F}_t\right].$$

The effect of credit risk is to replace the short rate r_s by an adjusted rate $r_s + \lambda_s$.

8.2 Forward Rate

8.2.1 Standard Forward Rate

The forward rate $f(t,T,S)$ is the rate agreed at time t for a loan over $[T,S]$, defined by exponential compounding as

$$f(t,T,S) = -\frac{\log P(t,S) - \log P(t,T)}{S - T}.$$

It is constructed by a two-step replication : borrow \$1 with maturity S and invest proceeds in a bond with maturity T .

8.2.2 Instantaneous Forward Rate

Instantaneous forward rate is the slope of the discount curves, it tell the value of 1 at time t when maturity is T. Taking $S \downarrow T$ gives the instantaneous forward rate

$$f(t,T) = -\frac{\partial \log P(t,T)}{\partial T},$$

which encodes the full term structure. Inverting this relation,

$$P(t,T) = \exp\left(-\int_t^T f(t,s) \, ds\right).$$

The short rate is recovered as the limit $r_t = \lim_{T\downarrow t} f(t,T)$.

8.2.3 LIBOR Forward Rate

LIBOR(London Inter Bank Offered Rate) is the interest rate at which a bank can borrow from another bank without collateral. Interest for consumers = LIBOR + spread. On the LIBOR market, linear (simple) compounding replaces exponential compounding :

$$L(t,T,S) = \frac{1}{S - T} \left(\frac{P(t,T)}{P(t,S)} - 1 \right).$$

LIBOR rates are always positive (as they are ratios of bond prices minus one divided by tenor length). The instantaneous LIBOR rate satisfies $r_t = L(t,t)$.

Key identity. For a tenor structure $\{T_i\}$,

$$P(t,T_j) = P(t,T_i) \prod_{k=i}^{j-1} \frac{1}{1 + \delta_k L(t,T_k,T_{k+1})},$$

where $\delta_k = T_{k+1} - T_k$.

8.3 Caps and Floors

8.3.1 Caplet

A caplet protects a floating-rate borrower against the spot forward rate $f(T,T,S)$ rising above a strike κ . Its payoff at time S is $(f(T,T,S) - \kappa)^+$.

Priced at time $t \leq T$ using the forward measure P^S with maturity S :

Caplet price = $\mathbb{E}\left[e^{-\int_t^S r_s \, ds} (f(T,T,S) - \kappa)^+ \mid \mathcal{F}_t\right] = P(t,S) \mathbb{E}^S[(f(T,T,S) - \kappa)^+ \mid \mathcal{F}_t]$.

The numeraire here is $P(t,S)$ and the change from Q to P^S removes the stochastic discount factor from the expectation.

8.3.2 Cap

A cap is a portfolio of caplets over a tenor structure $\{T_i, \dots, T_j\}$:

$$\text{Cap price} = \sum_{k=i}^{j-1} \delta_k \, P(t,T_{k+1}) \, \mathbb{E}^{k+1}[(f(T_k,T_k,T_{k+1}) - \kappa)^+ \mid \mathcal{F}_t].$$

Each caplet is priced under a different forward measure P^{k+1} .

8.3.3 Floor

A floor is the put counterpart to a cap ; it protects a lender against rates falling below κ . Floor payoff : $(\kappa - f(T,T,S))^+$.

8.4 Swap

An interest rate swap exchanges a floating forward rate $f(t,T,S)$ against a fixed rate κ , generating net cash flow

$$\sum_{k=i}^{j-1} \delta_k P(t,T_{k+1}) (f(t,T_k,T_{k+1}) - \kappa).$$

Swap rate. The fair fixed rate $S(t,T_i,T_j)$ that makes this cash flow zero is called the forward swap rate :

$$S(t,T_i,T_j) = \frac{P(t,T_i) - P(t,T_j)}{P(t,T_i,T_j)},$$

where the annuity numeraire (present value of the tenor) is

$$P(t,T_i,T_j) = \sum_{k=i}^{j-1} \delta_k P(t,T_{k+1}).$$

This clean formula for S in terms of bond prices holds only for LIBOR-style (simple compounding) rates.

8.5 Swaption

A swaption is an option on the above swap. The payer swaption has payoff at T_i :

$$\left(\sum_{k=i}^{j-1} \delta_k P(T_i,T_{k+1})(L(T_i,T_k,T_{k+1}) - \kappa)\right)^+ = P(T_i,T_i,T_j)(S(T_i,T_i,T_j) - \kappa)^+.$$

This simplification (decomposing the sum into the swap rate times the annuity) is essential for tractable pricing.

Priced at $t \leq T_i$ using the forward swap measure $P^{i,j}$:

$$\text{Swaption price} = P(t,T_i,T_j) \, \mathbb{E}^{i,j}[(S(T_i,T_i,T_j) - \kappa)^+ \mid \mathcal{F}_t].$$

8.6 Bond Option

A European call option on the bond $P(T,S)$ with strike K has payoff $(P(T,S) - K)^+$ and price

$$\mathbb{E}\left[e^{-\int_t^T r_s \, ds} (P(T,S) - K)^+ \mid \mathcal{F}_t\right] = P(t,T) \, \mathbb{E}^{\tilde{P}}[(P(T,S) - K)^+ \mid \mathcal{F}_t].$$

In the Vasicek model, the analytical formula is

$$= P(t,S) \Phi\left(\frac{v}{2} + \frac{1}{v} \log \frac{P(t,S)}{K P(t,T)}\right) - K P(t,T) \Phi\left(-\frac{v}{2} + \frac{1}{v} \log \frac{P(t,S)}{K P(t,T)}\right),$$

where $v^2 = \frac{\sigma^2}{2b^2} C^2 (S - T)(1 - e^{-2b(T-t)})$ and Φ is the Gaussian cdf. This is a Black-Scholes-type formula with $P(t,S)/P(t,T)$ playing the role of the “forward price” of the bond.

8.7 Credit Default Swap (CDS)

A CDS transfers credit risk from buyer to seller :

- Premium leg** : buyer pays fixed spread S_t at times $T_{i+1}, \dots, T_j$ as long as no default has occurred. Value :

$$V(t,T) = S_t \, P(t,T_i,T_j)$$

where $P(t,T_k)$ are defaultable bond prices.

- Protection leg** : seller pays $(1 - \xi_{k+1})$ upon default at T_{k+1} .

The fair spread equates the two legs :

$$S_t = \frac{(1 - \xi) \sum_{k=i}^{j-1} \mathbb{E} \left[\mathbf{1}_{(T_k, T_{k+1}]}(\tau) e^{-\int_t^{T_{k+1}} r_s ds} \mid \mathcal{G}_t \right]}{P(t, T_i, T_j)}.$$

9 Short Rate Models

9.1 Vasicek Model

The Vasicek (1977) model is the workhorse for explicit calculations :

$$dr_t = (a - br_t) dt + \sigma dB_t.$$

Properties : mean-reversion to a/b , Gaussian invariant distribution $\mathcal{N}(a/b, \sigma^2/(2b))$, allows negative rates.

Explicit solution (from the Ornstein-Uhlenbeck SDE) :

$$r_t = r_0 e^{-bt} + \frac{a}{b}(1 - e^{-bt}) + \sigma \int_0^t e^{-b(t-s)} dB_s.$$

Bond price. Using the affine ansatz $P(t, T) = e^{A(T-t) + C(T-t)r_t}$:

$$C(s) = -\frac{1}{b}(1 - e^{-bs}), \quad A(s) = \frac{4ab - 3\sigma^2}{4b^3} + \frac{\sigma^2 - 2ab}{2b^2}s + \frac{\sigma^2 - ab}{b^3}e^{-bs} - \frac{\sigma^2}{4b^3}e^{-2bs}.$$

9.2 Cox-Ingersoll-Ross (CIR) Model

$$dr_t = (\alpha - \beta r_t) dt + \sigma \sqrt{r_t} dB_t.$$

The square-root diffusion coefficient guarantees $r_t \geq 0$ when $2\alpha \geq \sigma^2$. Bond prices are of the affine form $e^{A - r_t B}$ with Riccati equations for A and B .

9.3 Hull-White Model

Time-dependent extension of Vasicek :

$$dr_t = (\theta(t) - a(t)r_t) dt + \sigma(t) dB_t.$$

The function $\theta(t)$ is calibrated to match any initial forward curve $f^M(0, T)$:

$$\theta(t) = a f^M(0, t) + \frac{\partial f^M}{\partial t}(0, t) + \frac{\sigma^2}{2a}(1 - e^{-2at}).$$

9.4 Two-Factor Model

To break the perfect correlation between bond prices of different maturities (a defect of one-factor affine models), one uses :

$$r_t = \varphi(t) + X_t + Y_t, \quad dX_t = -aX_t dt + \sigma dB_t^{(1)}, \quad dY_t = -bY_t dt + \eta dB_t^{(2)},$$

where $dB_t^{(1)} dB_t^{(2)} = \rho dt$. The bond price has the explicit form

$$P(t, T) = e^{-\int_t^T \varphi(s) ds} \cdot e^{-\frac{1-e^{-a(T-t)}}{a} X_t} \cdot e^{-\frac{1-e^{-b(T-t)}}{b} Y_t} \cdot e^{(\text{variance and covariance terms})}.$$

10 Forward Rate Models

10.1 The HJM Framework

Heath, Jarrow and Morton (1992) model the entire forward curve directly. The instantaneous forward rate satisfies the SDE

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dB_t.$$

HJM absence-of-arbitrage condition. The discounted bond price

$e^{-\int_0^t r_s ds} P(t, T)$ must be a martingale. Differentiating with respect to T yields the constraint

$$\alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, s) ds.$$

The drift of the forward rate is entirely determined by its own volatility structure. This is the fundamental result of the HJM framework.

Markov property. The short rate $r_t = f(t, t)$ is Markovian in the HJM model if and only if the volatility has the product form $\sigma(s, t) = \xi(s)\psi(t)$. Under this condition,

$$\mathbb{E}[Z_t \mid \mathcal{F}_u] = \frac{\psi(t)}{\psi(u)} Z_u, \quad Z_t = \int_0^t \xi(s) \psi(s) dB_s.$$

The Vasicek model satisfies this with $\sigma(s, t) = \sigma e^{-b(t-s)}$.

10.2 BGM (Brace-Gatarek-Musiela) Model

The BGM model specifies dynamics for LIBOR rates $L(t, T_i, T_{i+1})$ directly, ensuring positivity. Under the forward measure P^{i+1} :

$$\frac{dL(t, T_i, T_{i+1})}{L(t, T_i, T_{i+1})} = \gamma_i(t) dB_t^{i+1},$$

so L is a geometric Brownian motion under P^{i+1} . Under P^n (the terminal measure) :

$$\frac{dL(t, T_i, T_{i+1})}{L(t, T_i, T_{i+1})} = -\gamma_i(t) \sum_{j=i+1}^{n-1} \frac{\delta_j \gamma_j(t) L(t, T_j, T_{j+1})}{1 + \delta_j L(t, T_j, T_{j+1})} dt + \gamma_i(t) dB_t^n.$$

The nonlinear drift arises from the change of measure between forward measures. The relation between bond volatilities and LIBOR volatilities is

$$\zeta_{i+1}(t) - \zeta_i(t) = -\frac{\delta_i L(t, T_i, T_{i+1}) \gamma_i(t)}{1 + \delta_i L(t, T_i, T_{i+1})},$$

derived by matching two expressions for $d(P(t, T_i)/P(t, T_{i+1}))$.

11 Change of Measure and Numeraire Methods

11.1 Risk-Neutral Measure

The Girsanov theorem transforms the real-world measure P to the risk-neutral measure Q under which discounted asset prices are martingales. The density is

$$\frac{dQ}{dP} = \exp \left(-\int_0^T \psi_s dB_s - \frac{1}{2} \int_0^T |\psi_s|^2 ds \right),$$

where $\psi_t = (\mu_t - r_t)/\sigma_t$ is the market price of risk. Under Q , $\hat{B}_t = B_t + \int_0^t \psi_s ds$ is a standard Brownian motion.

11.2 Forward Measure

For each maturity T , the forward measure $\tilde{P}$ (or P^T) is defined by

$$\frac{d\tilde{P}}{dP} = \frac{1}{P(0, T)} e^{-\int_0^T r_s ds}.$$

The conditional density evolves as

$$\frac{d\tilde{P}|_{\mathcal{F}_t}}{dP|_{\mathcal{F}_t}} = \frac{e^{-\int_t^T r_s ds}}{P(t, T)}.$$

Key pricing formula :

$$\mathbb{E} \left[F e^{-\int_t^T r_s ds} \mid \mathcal{F}_t \right] = P(t, T) \mathbb{E}^{\tilde{P}}[F \mid \mathcal{F}_t].$$

This separates the stochastic discount factor from the payoff, enabling pricing via an expectation that does not involve the integral $\int r ds$.

Brownian motion under forward measure. If

$dP(t, T)/P(t, T) = r_t dt + \zeta_t dB_t$, then

$$\tilde{B}_t = B_t - \int_0^t \zeta_s ds$$

is a standard Brownian motion under $\tilde{P}$. This is a direct application of the Girsanov theorem with shift $-\zeta$.

Martingale under forward measure. The ratio $P(t, S)/P(t, T)$ is a martingale under P^T . This is used critically :

$$\mathbb{E}^{\tilde{P}}[P(T, S) \mid \mathcal{F}_t] = \frac{P(t, S)}{P(t, T)},$$

i.e., the current forward price of the bond is its expected future price under the T -forward measure.

11.3 Multiple Forward Measures

For a tenor structure $\{T_0, \dots, T_n\}$, the i -th forward measure P^i is defined by

$$\frac{dP^i}{dP} = \frac{e^{-\int_0^{T_i} r_s ds}}{P(0, T_i)}, \quad \frac{dP^i|_{\mathcal{F}_t}}{dP|_{\mathcal{F}_t}} = \frac{e^{-\int_t^{T_i} r_s ds}}{P(t, T_i)}.$$

Under P^i , the process $B_t^i = B_t - \int_0^t \zeta_i(s) ds$ is a standard Brownian motion. The ratio $P(t, T_j)/P(t, T_i)$ is a martingale under P^i .

Change between measures :

$$dB_t^j = dB_t^i - (\zeta_j(t) - \zeta_i(t)) dt.$$

11.4 Forward Swap Measure

For a swaption on $[T_i, T_j]$, the natural numeraire is the annuity $P(t, T_i, T_j)$. The forward swap measure $P^{i,j}$ is defined by

$$\frac{dP^{i,j}}{dP} = e^{-\int_0^{T_i} r_s ds} \frac{P(T_i, T_i, T_j)}{P(0, T_i, T_j)}.$$

Under $P^{i,j}$, the swap rate $S(t, T_i, T_j)$ is a martingale. The process

$$B_t^{i,j} = B_t - \sum_{l=i}^{j-1} \delta_l \int_0^t v_{l+1}^{i,j}(s) \zeta_{l+1}(s) ds$$

is a standard Brownian motion under $P^{i,j}$, where $v_k^{i,j}(t) = P(t, T_k)/P(t, T_i, T_j)$.

12 Martingale and Markov Properties

12.1 Martingale Property in Bond Pricing

The central martingale in interest rate theory is the discounted bond price :

$$t \mapsto e^{-\int_0^t r_s ds} P(t, T) = \mathbb{E}^Q \left[e^{-\int_0^T r_s ds} \mid \mathcal{F}_t \right].$$

This is a martingale under Q by the tower property of conditional expectations. The fact that all terms in dt vanish when we apply Ito's formula to this process gives us the bond pricing PDE.

How it is used : Write $P(t, T) = F(t, r_t)$, apply Ito's formula to $e^{-\int_0^t r_s ds} F(t, r_t)$. Since this is a martingale, the finite variation (dt) terms must vanish, yielding the PDE

$$xF(t, x) = \tilde{\mu}(t, x) \frac{\partial F}{\partial x} + \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 F}{\partial x^2} + \frac{\partial F}{\partial t}.$$

12.2 Markov Property

All solutions of SDEs have the Markov property. This means

$$\mathbb{E}[f(r_{t_1}, \dots, r_{t_n}) \mid \mathcal{F}_t] = \mathbb{E}[f(r_{t_1}, \dots, r_{t_n}) \mid r_t],$$

so $P(t, T)$ depends only on r_t (not the entire history), making it a function $F(t, r_t)$. This dimensional reduction from an infinite-dimensional object to a single state variable is the key step enabling PDE methods.

In the two-factor model, (X_t, Y_t) is jointly Markovian, so $P(t, T) = F(t, X_t, Y_t)$ depends on two state variables and satisfies a 2D PDE.

12.3 Martingale Property of Swap Rate

Under $P^{i,j}$, the swap rate $S(t, T_i, T_j)$ is a martingale because it equals $v_i^{i,j}(t) - v_j^{i,j}(t)$, and each $v_k^{i,j}(t) = P(t, T_k)/P(t, T_i, T_j)$ is a martingale under $P^{i,j}$ by a ratio-of-martingales argument.

Its dynamics under $P^{i,j}$ are

$$dS(t, T_i, T_j) = S(t, T_i, T_j) \sigma_{i,j}(t) dB_t^{i,j},$$

which is the key fact enabling Black-Scholes-type swaption pricing.

13 Pricing Techniques

13.1 PDE Method

Given $P(t, T) = F(t, r_t)$, apply Ito's formula to derive :

- The drift must vanish (martingale condition), giving a PDE.
- In the Vasicek model, the affine guess $F = e^{A(T-t) + C(T-t)r}$ reduces the PDE to two ODEs (Riccati + linear). This is solvable in closed form.

13.2 Probabilistic Method

Compute $P(t, T) = \mathbb{E}^Q[e^{-\int_t^T r_s ds} \mid \mathcal{F}_t]$ directly. In the Vasicek model, $r_s = g(s) + \int_0^s h(s, u) dB_u$ is Gaussian, so $\int_t^T r_s ds$ is also Gaussian given $\mathcal{F}_t$. Using the Laplace transform of a Gaussian :

$$\mathbb{E}[e^{-X}] = e^{-\mathbb{E}[X] + \frac{1}{2} \text{Var}(X)},$$

all quantities reduce to computing the conditional mean and variance of $\int_t^T r_s ds$, which involve deterministic integrals.

13.3 Black-Scholes Formula

The classical formula $C(t, x) = \text{Bl}(K, x, \tilde{\sigma}_t, \tilde{r}_t, T - t)$ is fundamental :

$$\text{Bl}(K, x, \sigma, r, \tau) = x\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$

$$d_1 = \frac{\log(x/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Application via forward measure. Once the payoff is written as $(e^{m+X} - K)^+$ where X is Gaussian with variance v^2 given $\mathcal{F}_t$, Lemma 2.3 gives :

$$\mathbb{E}[(e^{m+X} - K)^+] = e^{m+v^2/2}\Phi(v + (m - \log K)/v) - K\Phi((m - \log K)/v).$$

This same formula is reused for caplets, bond options, and swaptions.

13.4 Cap Pricing in BGM

Since $L(t, T_i, T_{i+1})$ is geometric Brownian motion under P^{i+1} :

$$\text{Caplet}(t) = P(t, T_{i+1}) \cdot \text{Bl}(\kappa, L(t, T_i, T_{i+1}), \sigma_i(t), 0, T_i - t),$$

where $\sigma_i(t)^2 = \frac{1}{T_i - t} \int_t^{T_i} |\gamma_i(s)|^2 ds$. This is a direct application of Black-Scholes under the forward measure, enabled by the BGM assumption that L is lognormal under P^{i+1} .

13.5 Swaption Pricing in BGM – Approximation

The swap rate volatility under $P^{i,j}$ is

$$\sigma_{i,j}^2(t) = \frac{1}{S(t, T_i, T_j)^2} \sum_{l,k=i}^{j-1} \gamma_l(t) \gamma_k(t) w_l^{i,j}(t) w_k^{i,j}(t) L(t, T_l, T_{l+1}) L(t, T_k, T_{k+1}),$$

with weights $w_k^{i,j}$. This is exact but involves stochastic coefficients.

Freezing approximation. Replace all random quantities at time t :

$$|\tilde{\sigma}_{i,j}(t)|^2 \approx \frac{1}{T_i - t} \sum_{l,k=i}^{j-1} \frac{\delta_k \delta_l v_{l+1}^{i,j}(t) v_{k+1}^{i,j}(t) L(t, T_l, T_{l+1}) L(t, T_k, T_{k+1})}{S(t, T_i, T_j)^2} \int_t^{T_i} \gamma_l(s) \gamma_k(s) ds$$

Then treat $S(t, T_i, T_j)$ as geometric Brownian motion with constant volatility $\tilde{\sigma}_{i,j}$, apply Black-Scholes :

$$\text{Swaption price} = P(t, T_i, T_j) \cdot \text{Bl}(\kappa, S(t, T_i, T_j), \tilde{\sigma}_{i,j}(t), 0, T_i - t).$$

13.6 BGM Calibration

Rebonato parametrization :

$$g_i(t) = |\gamma_i|(t) = g_\infty + (1 + a(T_i - t) - g_\infty)e^{-b(T_i - t)}.$$

Parameters a, b, g_∞ are estimated by minimizing the root-mean-square distance between market swaption volatilities $\sigma_{i,j}^B$ and model volatilities $\sigma_{i,j}(t, b, g_\infty)$:

$$\text{RMS}(b, g_\infty) = \sqrt{\frac{2}{(n-1)(n-2)} \sum_{i,j} \left(\frac{\sigma_{i,j}^B - \sigma_{i,j}}{\sigma_{i,j}^B} \right)^2}.$$

14 Hedging Strategies

14.1 Delta Hedging

For a European option $F = \phi(S_T)$, the hedging portfolio requires $\eta_t = \partial C / \partial x(t, S_t)$ units of the risky asset (the Delta). Under the self-financing condition $dV_t = \zeta_t dA_t + \eta_t dS_t$, the portfolio replicates F exactly.

14.2 Bond Option Hedging

To hedge $(P(T, S) - \kappa)^+$ using bonds with maturities T and S :

$$\xi_t^S = \Phi\left(\frac{\log(X_t/\kappa)}{v(t, T)} + \frac{v(t, T)}{2}\right), \quad \xi_t^T = -\kappa\Phi\left(\frac{\log(X_t/\kappa)}{v(t, T)} - \frac{v(t, T)}{2}\right),$$

where $X_t = P(t, S)/P(t, T)$ is the forward bond price and

$$v^2(t, T) = \int_t^T |\sigma_s^S - \sigma_s^T|^2 ds.$$

The key technique is **numeraire invariance** : the forward portfolio $\hat{V}_t = V_t/P(t, T)$ satisfies $d\hat{V}_t = \frac{\partial C}{\partial x} dX_t$, reducing the hedging problem to a standard Black-Scholes delta hedge in the forward measure.

14.3 Swaption Hedging

For a swaption on $[T_i, T_j]$ with payoff $(S(T_i, T_i, T_j) - \kappa)^+$, the hedge uses bonds of all maturities $T_i, \dots, T_j$:

$$\xi_t^i = \Phi\left(\frac{\log(S_t/\kappa)}{v(t, T_i)} + \frac{v(t, T_i)}{2}\right), \quad \xi_t^j = -\xi_t^i - \kappa\delta_{j-1}\Phi\left(\frac{\log(S_t/\kappa)}{v(t, T_i)} - \frac{v(t, T_i)}{2}\right),$$

with intermediate maturities hedging the annuity :

$$\xi_t^k = -\kappa\delta_{k-1}\Phi\left(\frac{\log(S_t/\kappa)}{v(t, T_i)} - \frac{v(t, T_i)}{2}\right), \quad i + 1 \leq k \leq j - 1.$$

15 Forward Rate Parametrizations

Nelson-Siegel : $g(x) = z_1 + (z_2 + z_3x)e^{-xz_4}$ (4 parameters, one hump).

Svensson : $g(x) = z_1 + (z_2 + z_3x)e^{-xz_4} + z_5xe^{-xz_6}$ (6 parameters, two humps).

These provide flexible shapes for forward curves but are shown to be inconsistent with the HJM no-arbitrage condition : HJM forward curves cannot remain in the Nelson-Siegel or Svensson spaces. This motivates the shift approach : use a deterministic initial curve $\varphi(T)$ (calibrated to market) plus a stochastic part from the Vasicek model.

16 Credit Risk Techniques

Survival probability. With hazard rate $\lambda(t)$:

$$P(\tau > T \mid \tau > t) = \exp\left(-\int_t^T \lambda(u) du\right).$$

Key lemma (Guo-Jarrow-Menn). For any $\mathcal{F}_T$ -measurable F :

$$\mathbb{E}[F\mathbf{1}_{\{\tau>T\}} \mid \mathcal{G}_t] = \mathbf{1}_{\{\tau>t\}}\mathbb{E}\left[F\exp\left(-\int_t^T \lambda_u du\right) \middle| \mathcal{F}_t\right].$$

This reduces defaultable bond pricing to standard bond pricing with shifted rate $r_s + \lambda_s$, using only the background filtration $\mathcal{F}_t$ instead of the enlarged filtration $\mathcal{G}_t$.

Survival probability under forward measure :

$$Q(t, T) = \mathbb{E}^{\tilde{P}}\left[\exp\left(-\int_t^T \lambda_s ds\right) \middle| \mathcal{F}_t\right].$$

The defaultable bond price factors as

$P(t, T) = \mathbf{1}_{\{\tau>t\}} \mathbb{E}[e^{-\int^r} | \mathcal{F}_t] \cdot Q(t, T) \cdot e^{U(t, T)}$ where U captures the correlation between r and λ .

17 Summary of Numeraire Choices

Instrument	Numeraire	Measure
Generic claim with payoff at T	$P(t, T)$	Forward measure P^T
Caplet with payoff at S	$P(t, S)$	Forward measure P^S
Bond option on $P(T, S)$	$P(t, T)$	Forward measure P^T
Swaption on $[T_i, T_j]$	$P(t, T_i, T_j)$	Swap measure $P^{i,j}$
BGM cap pricing	$P(t, T_{i+1})$	Forward measure P^{i+1}
Self-financing portfolio	bank account A_t	Risk-neutral Q

The guiding principle : choose the numeraire to be the asset that appears in the payoff denominator, so that the normalized payoff is a martingale under the corresponding measure. This eliminates the stochastic discount factor from the pricing formula.